Robust filters for UAV navigation

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Motivation: Early Detection and Rescue of Wildfire

a)

b)

The forest fires: a) in California now!!!
b) in Chongqing in August 2022

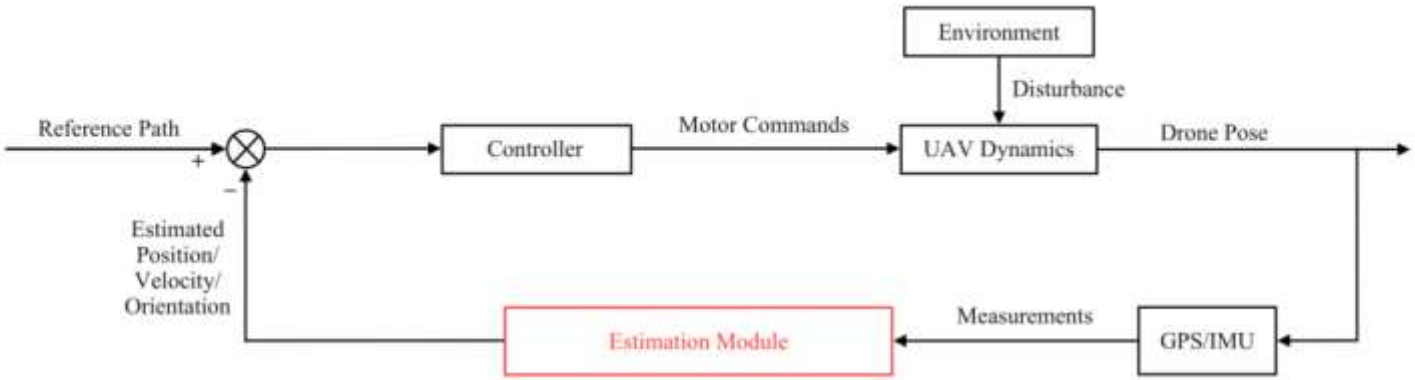
- Thousands of firefighters
- High-risk work
- Many firefighters injured

- Operator fatigue
- Low visibility
- Wide coverage area
- Geographic location

Firefighters fighting

Unmanned aerial vehicle system

State of Art: UAV Pose Estimation



UAV navigation system

GPS (Global positioning system)

- Reliable
- Subject to the environment
- Low sampling frequency

IMU (Inertial measurement unit)

- Subject to the cumulative error
- Independent from the environment
- High sampling frequency

Aided Inertial Navigation Model

- Process model:

$$\begin{bmatrix} \dot{q} \\ \dot{v} \\ \dot{p} \\ \dot{\omega}_b \\ \dot{a}_b \end{bmatrix} = \begin{bmatrix} \frac{1}{2} q * (\omega_m - \omega_b) + q * \omega_n \\ g_N + q * (a_m - a_b) * q^{-1} + q * v_n * q^{-1} \\ v \\ 0 \\ 0 \end{bmatrix}$$

  - State: $[q, v, p, w_b, a_b]^T$
  - Input: $[w_m,\ w_b]^T$

- Measurement model:

$y_k = Cx_k + \epsilon_k$ where $y = [v_n,\ v_e,\ v_d,\ p_n,\ p_e,\ p_d]^T$, and $C = \begin{bmatrix} 0 & I_3 & 0 & 0 & 0 \\ 0 & 0 & I_3 & 0 & 0 \end{bmatrix}$.

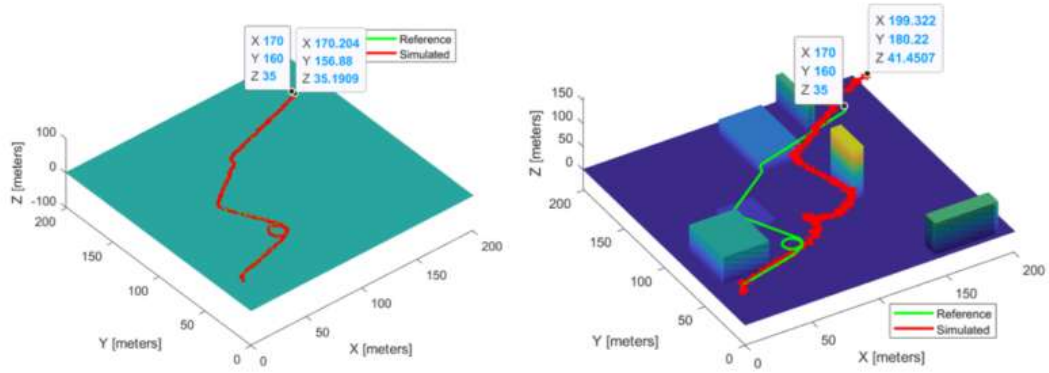
  - Measurements (y_k): A GPS receiver and a barometer that gather the position measurements as well as the corresponding velocities in North-East-Down frame (n-frame), and ϵ_k is the zero mean white Gaussian noise with variance R.

Aided Inertial Navigation Model

- State:
- Unit quaternion: $q = \begin{bmatrix} q_0 & q_1 & q_2 & q_3 \end{bmatrix}^\top \in \mathbb{R}^4$ representing the orientation of the body-fixed frame (b-frame) with respect to n-frame;
- Velocity vector: $v \in \mathbb{R}^3$ [m/s] in n-frame;
- Position vector: $p \in \mathbb{R}^3$ [m] in n-frame;
- Constant vector bias on gyroscopes: $\omega_b \in \mathbb{R}^3$ [rad/s] in b-frame;
- Constant vector bias on accelerometers: $a_b \in \mathbb{R}^3$ [m/s^2] in b-frame;
- Input:
- Instantaneous angular velocity vector: $\omega_m \in \mathbb{R}^3$ [rad/s] measured by the gyroscopes;
- Specific acceleration vector: $a_m \in \mathbb{R}^3$ [m/s^2] measured by the accelerometers;

State of Art: UAV Pose Estimation

Chongqing City



a. Obstacle-free environment

b. GPS-denied environment

Simulated trajectories of UAV in the different environments

GPS signals could be denied!!!

We need more **External Sensing, e.g. Optical flow sensor**

UAV Description

Table 1. Equipment

MTF-01P Optical Flow&Range Sensor	**https://micoair.com/optical_range_sensor_mtf-01p/**
Holybro M8N GPS	https://holybro.com/collections/gps/products/m8n-gps
PX4 Autopilot	https://px4.io/

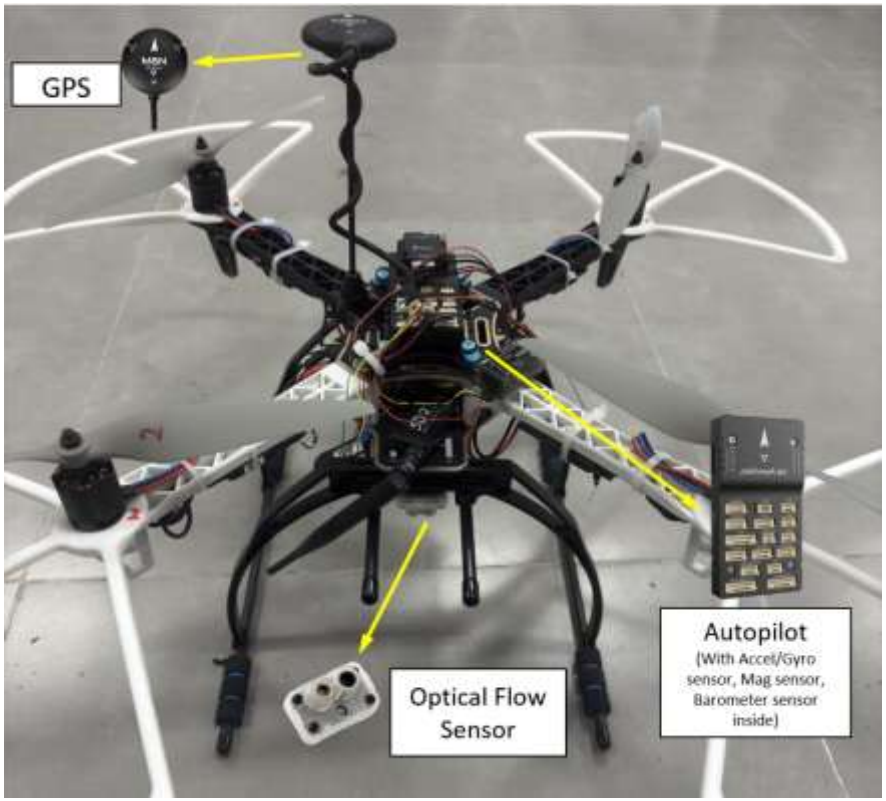


Table 2. Sensor details in Autopilot

On-board Sensor	Description
Accel/Gyro1	ICM-42688-P
Accel/Gyro2	BMI055
Mag	IST8310
Barometer	MS5611

Experiment Description

Test site

Drone aerial view

Satellite image from Google map:
40.15952N, 116.29353W

Flight description:
The desired flight speed between the planned points is set to 1.5m/s. The planned path and the final actual flight trajectory are shown below, respectively:

REKF

Algorithm 1 Robust Extended Kalman filter at time t

Input $\hat{x}_t$, V_t, c_t, y_t, u_t

Output $\hat{x}_{t|t}$, $\hat{x}_{t+1}$, V_{t+1}

$C_t = \left.\frac{\partial h(x,u_t)}{\partial x}\right|_{x=\hat{x}_t}$

$L_t = V_t C_t^T (C_t V_t C_t^T + DD^T)^{-1}$

$\hat{x}_{t|t} = \hat{x}_t + L_t[y_t - h(\hat{x}_t, u_t)]$

$A_t = \left.\frac{\partial f(x,u_t)}{\partial x}\right|_{x=\hat{x}_{t|t}}$

$\hat{x}_{t+1} = f(\hat{x}_{t|t}, u_t)$

$P_{t+1} = A_t V_t A_t^T$

$\quad -A_t V_t C_t^T (C_t V_t C_t^T + DD^T)^{-1} C_t V_t A_t^T + BB^T$

find θ_t *s.t.* $\gamma(P_{t+1}, \theta_t) = c_t$

$V_{t+1} = (P_{t+1}^{-1} - \theta_t I)^{-1}$

The larger c is, the greater the uncertainty is, in other words, the worse the measurement data is.

Try adjusting this parameter based on your experience!!!

Algorithm 1 Prediction resilient filter at time t

Input $\hat{x}_t$, $\tilde{P}_t$, c_t, y_t

Output $\hat{x}_{t+1}$, $\tilde{P}_{t+1}$, θ_t

$\mathcal{X}_t^i = \sigma_i(\hat{x}_t, \tilde{P}_t), \quad i = 1 \dots p$

$m_{y_t} = \sum_{i=1}^p W_m^i h(\mathcal{X}_t^i)$

$K_{y_t} = \sum_{i=1}^p W_c^i (h(\mathcal{X}_t^i) - m_{y_t})(h(\mathcal{X}_t^i) - m_{y_t})^\top + DD^\top$

$K_{x_t y_t} = \sum_{i=1}^p W_c^i (\mathcal{X}_t^i - \hat{x}_t)(h(\mathcal{X}_t^i) - m_{y_t})^\top$

$L_t = K_{x_t y_t} K_{y_t}^{-1}$

$\hat{x}_{t|t} = \hat{x}_t + L_t (y_t - m_{y_t})$

$P_{t|t} = \tilde{P}_t - L_t K_{y_t} L_t^\top$

$\mathcal{X}_{t|t}^i = \sigma_i(\hat{x}_{t|t}, P_{t|t}), \quad i = 1 \dots p$

$\hat{\mathcal{X}}_{t+1}^i = f(\mathcal{X}_{t|t}^i), \quad i = 1 \dots p$

$\hat{x}_{t+1} = \sum_{i=1}^p W_m^i \hat{\mathcal{X}}_{t+1}^i$

$P_{t+1} = \sum_{i=1}^p W_c^i (\hat{\mathcal{X}}_{t+1}^i - \hat{x}_{t+1})(\hat{\mathcal{X}}_{t+1}^i - \hat{x}_{t+1})^\top + BB^\top$

Find θ_t s.t. $\gamma(P_{t+1}, \theta_t) = c_t$

$\tilde{P}_{t+1} = (P_{t+1}^{-1} - \theta_t I)^{-1}$

- **Unscented Kalman filter (UKF).** The sigma points are obtained through the unscented transformation:

$$\sigma_i(\hat{x}, P) = \begin{cases} \hat{x} + \sqrt{\lambda + n}(\sqrt{P})_i, & \text{if } 1 \le i \le n \\ \hat{x} - \sqrt{\lambda + n}(\sqrt{P})_{i-n}, & \text{if } n+1 \le i \le 2n \\ \hat{x}, & \text{if } i = 2n+1 \end{cases}$$

where $(\sqrt{P})_i \in \mathbb{R}^n$ is the i-th column of $\sqrt{P}$ which is a square root matrix of P. The corresponding weights are

$$W_m^i = \lambda/(n+\lambda), \; W_c^i = W_m^i + 1 - a^2 + b, \quad \text{if } i = 2n+1$$
$$W_m^i = W_c^i = 1/(2(n+\lambda)), \quad \text{if } 1 \le i \le 2n \tag{5}$$

where $\lambda = a^2(\kappa + n) - n$, and the parameters a, b and κ can be chosen as suggested in [17].

Tasks

- **Find a suitable optical flow measurement model (non-linear) to replace the original GPS measurement model;**
- **Verify the accuracy of the provided measurement model based on the provided data (estimator_optical_flow, vehicle_optical_flow and sensor_optical_flow);**
- **Try to run EKF and UKF using your optical flow measurement model;**
- **Try to run REKF and RUKF using your optical flow measurement model;**

Thank you for your attention!

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